

- Message Logging
 - Log Info

```
// Update UI
function updatePetInfoInHtml() {
  $('.name').text(pet_info.name);
  $('.weight').text(pet_info.weight);
  $('.happiness').text(pet_info.happiness);
  $('.pet-image').attr('src', pet_info.image);

  console.info("Pet updated:", pet_info);
}
```

Live reload enabled.

index.html:81

Pet updated: ▼ Pet ⓘ

script.js:92

```
  happiness: 50
  image: "images/hound.jpg"
  name: "Sunika"
  weight: 50
  ▶ [[Prototype]]: Object
```

Pet updated:

script.js:92

```
▼ Pet {name: 'Sunika', weight: 52, happiness: 55, image: 'images/hound.jpg'} ⓘ
  happiness: 73
  image: "images/hound.jpg"
  name: "Sunika"
  weight: 51
  ▶ [[Prototype]]: Object
```

Pet updated:

script.js:92

```
▼ Pet {name: 'Sunika', weight: 51, happiness: 63, image: 'images/hound.jpg'} ⓘ
  happiness: 73
  image: "images/hound.jpg"
  name: "Sunika"
  weight: 51
  ▶ [[Prototype]]: Object
```

Pet updated:

script.js:92

```
▼ Pet {name: 'Sunika', weight: 51, happiness: 73, image: 'images/hound.jpg'} ⓘ
  happiness: 73
  image: "images/hound.jpg"
  name: "Sunika"
  weight: 51
  ▶ [[Prototype]]: Object
```

- Log Warning

```
99  function switchPet() {
100      currentPetIndex++;
101
102      if (currentPetIndex >= pets.length) {
103          console.warn("Reached end of pet list. Resetting index.");
104          currentPetIndex = 0;
105      }
106
107      pet_info = pets[currentPetIndex];
108
109      showMessage("Now playing with " + pet_info.name + " 🐾");
110      checkAndUpdatePetInfoInHtml();
111  }
```

```
Elements Console Sources Network >> 1 [Refresh] [Settings] [More] [Close]
[Icons] [Filter] top [Filter] Default levels No Issues [Settings]

Live reload enabled. index.html:81
Pet updated: ▾ Pet ⓘ script.js:94
  happiness: 50
  image: "images/hound.jpg"
  name: "Sunika"
  weight: 50
  ▸ [[Prototype]]: Object

Pet updated: script.js:94
  ▾ Pet {name: 'Helminth', weight: 40, happiness: 60, image: 'images/hound2.jpg'} ⓘ
    happiness: 60
    image: "images/hound2.jpg"
    name: "Helminth"
    weight: 40
    ▸ [[Prototype]]: Object

Pet updated: script.js:94
  ▾ Pet {name: 'Medjay', weight: 30, happiness: 80, image: 'images/hound3.jpg'} ⓘ
    happiness: 80
    image: "images/hound3.jpg"
    name: "Medjay"
    weight: 30
    ▸ [[Prototype]]: Object

⚠ ▸ Reached end of pet list. Resetting index. script.js:103

Pet updated: script.js:94
  ▾ Pet {name: 'Sunika', weight: 50, happiness: 50, image: 'images/hound.jpg'} ⓘ
    happiness: 50
    image: "images/hound.jpg"
    name: "Sunika"
    weight: 50
    ▸ [[Prototype]]: Object
```

- Log Error

```
74 // Prevent negative values
75 function checkWeightAndHappinessBeforeUpdating() {
76     if (pet_info.weight < 0) {
77         console.error("Weight dropped below 0. Resetting to 0.");
78         pet_info.weight = 0;
79     }
80
81     if (pet_info.happiness < 0) {
82         console.error("Happiness dropped below 0. Resetting to 0.");
83         pet_info.happiness = 0;
84     }
85 }
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 6, image: 'images/hound.jp
      g'}
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 3, image: 'images/hound.jp
      g'}
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 0, image: 'images/hound.jp
      g'}
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
```

```
✖ ▶ Happiness dropped below 0. Resetting to 0. script.js:82
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 0, image: 'images/hound.jp
      g'}
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
```

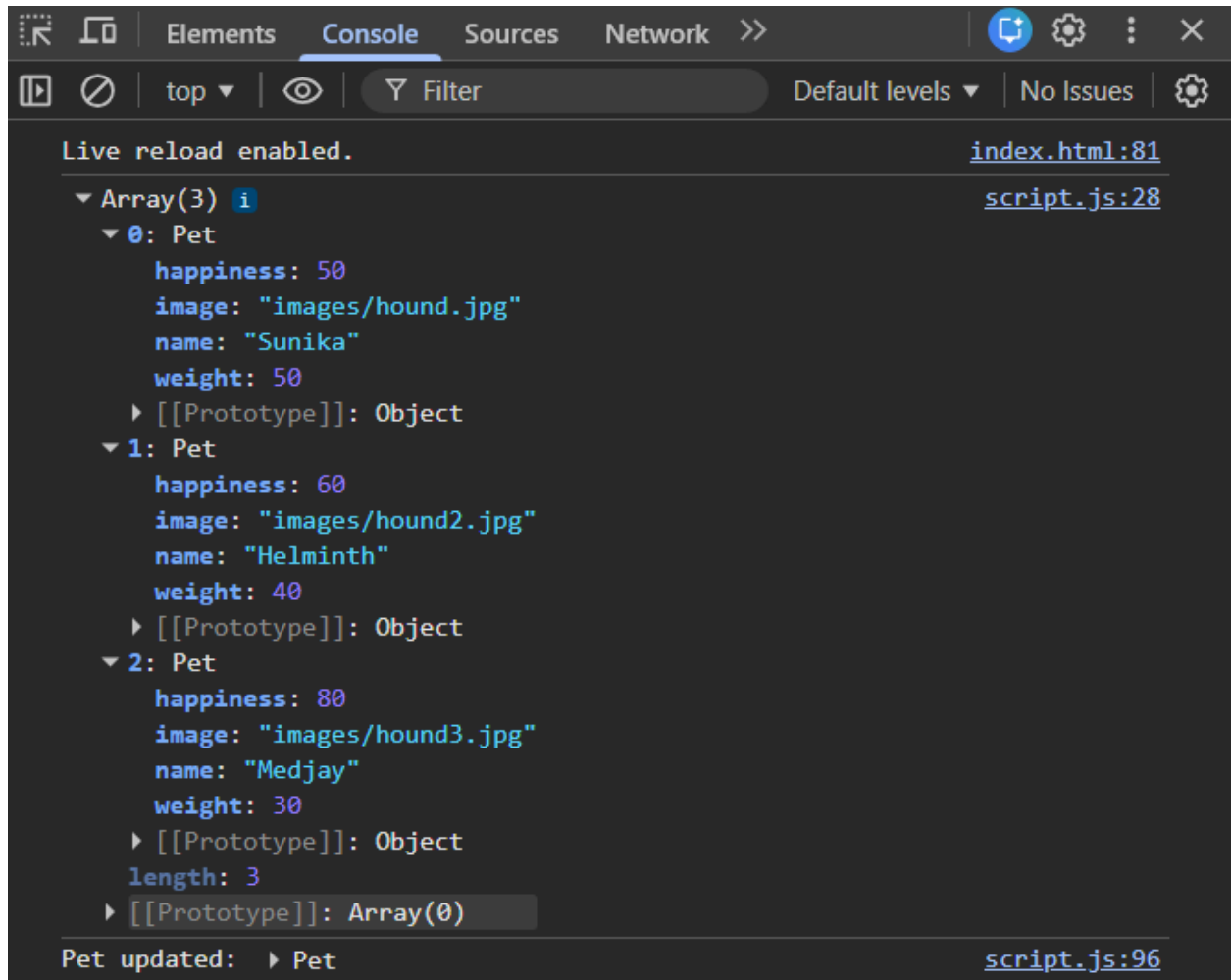
```
✖ ▶ Happiness dropped below 0. Resetting to 0. script.js:82
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 0, image: 'images/hound.jp
      g'}
```

```
✖ ▶ Weight dropped below 0. Resetting to 0. script.js:77
```

```
✖ ▶ Happiness dropped below 0. Resetting to 0. script.js:82
  Pet updated: script.js:94
    ▶ Pet {name: 'Sunika', weight: 0, happiness: 0, image: 'images/hound.jp
      g'}
```

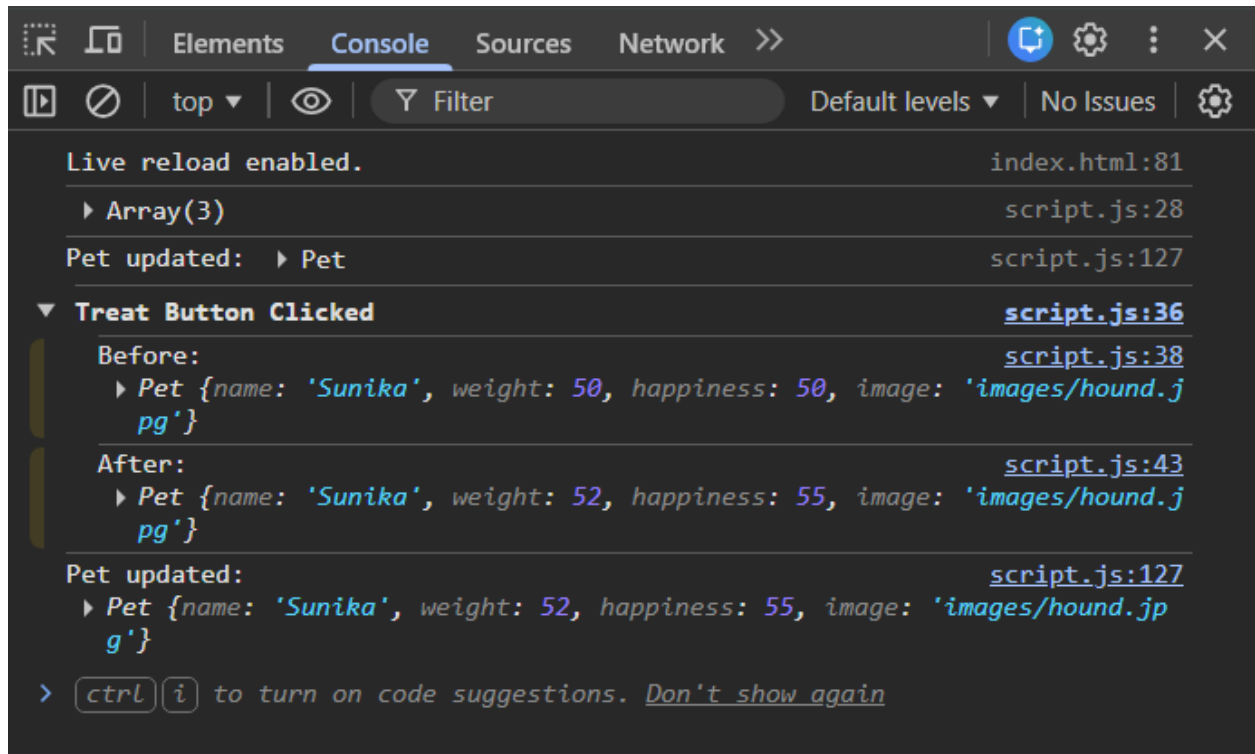
- Log Table

```
27
28 console.table(pets); // Log pet data in a table format for easy debugging
29
```



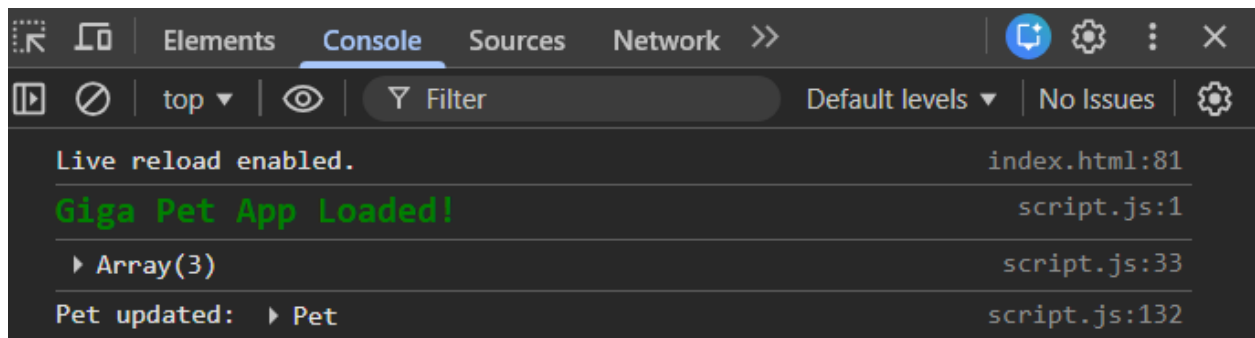
- Log Group

```
34 // Treat → +happiness +weight
35 function clickedTreatButton() {
36     console.group("Treat Button Clicked"); // Group logs for better organization
37
38     console.log("Before:", pet_info); // Log pet info before changes for debugging
39
40     pet_info.happiness += 5;
41     pet_info.weight += 2;
42
43     console.log("After:", pet_info); // Log pet info after changes to verify updates
44
45     console.groupEnd(); // End log group
46
47     showMessage("Yum! That was tasty! 🍖");
48     checkAndUpdatePetInfoInHtml();
49 }
```



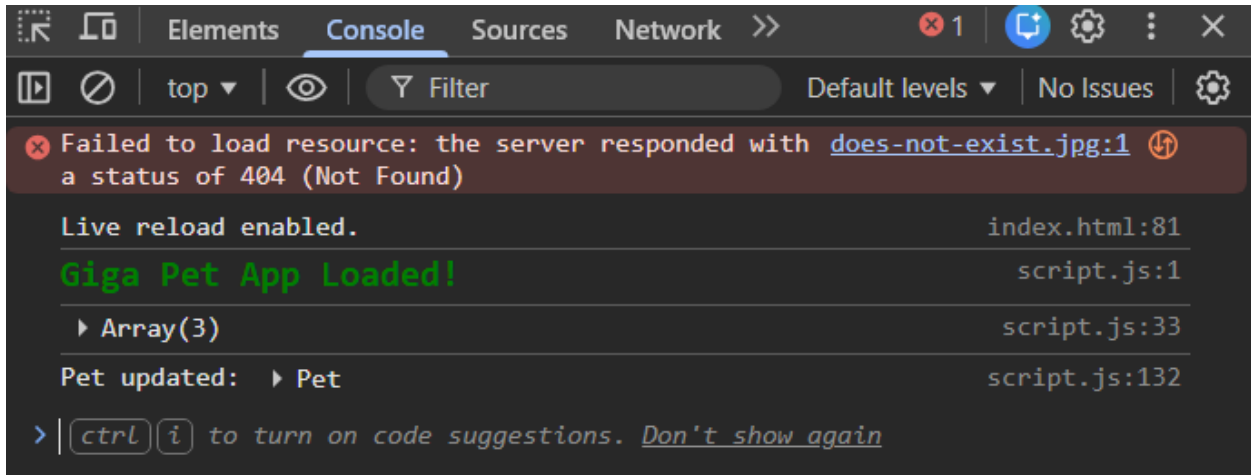
- Log Custom

```
1 console.log(  
2   "%cGiga Pet App Loaded!",  
3   "color: green; font-size: 16px; font-weight: bold;"  
4 ); // Initial load message with styling for better visibility
```

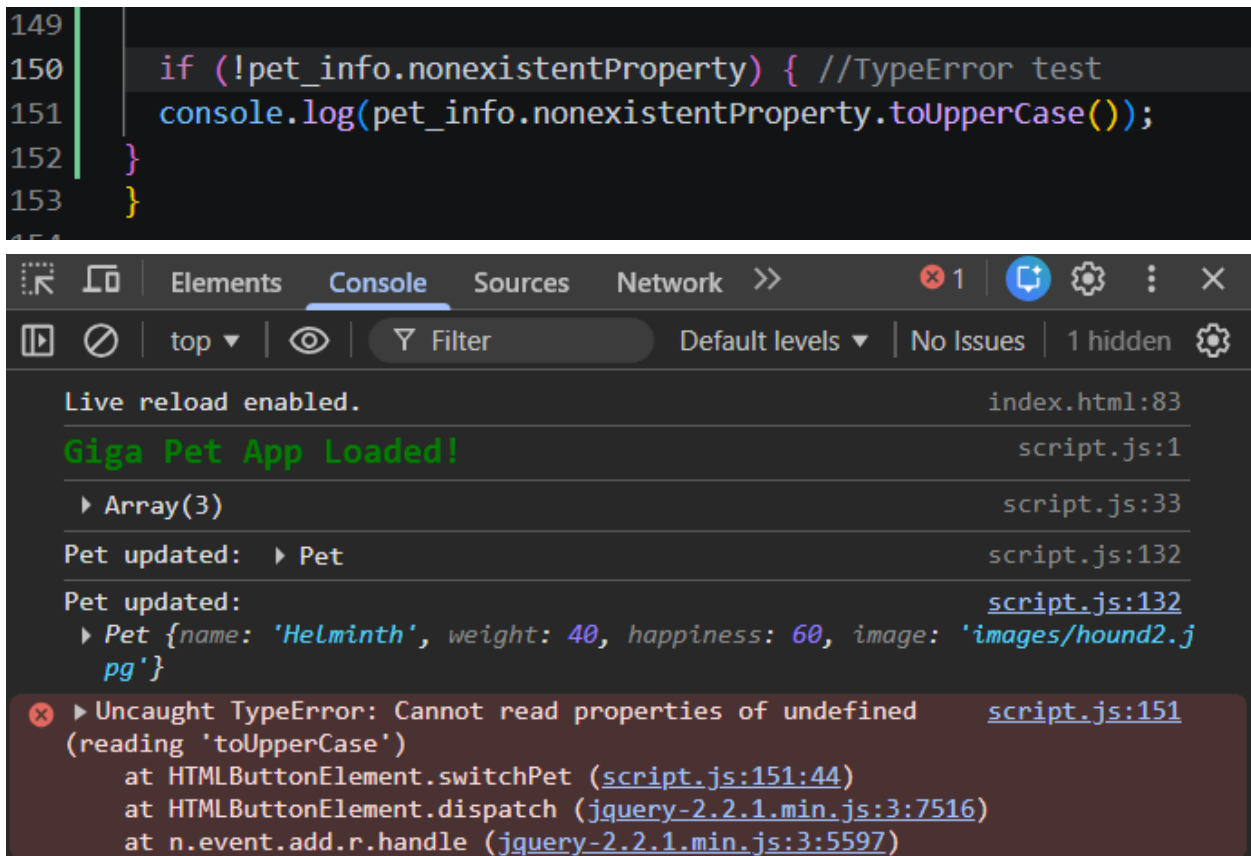


- View messages logged by the browser
 - Cause 404 network error

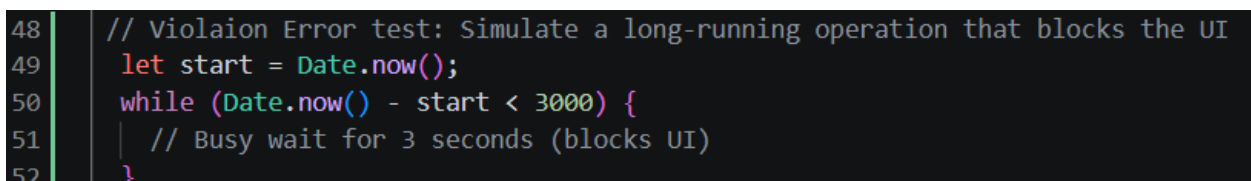
```
21 <section class="pet-image-container">  
22     
23   <!--   
24   <!-- 404 file test -->  
25 </section>
```

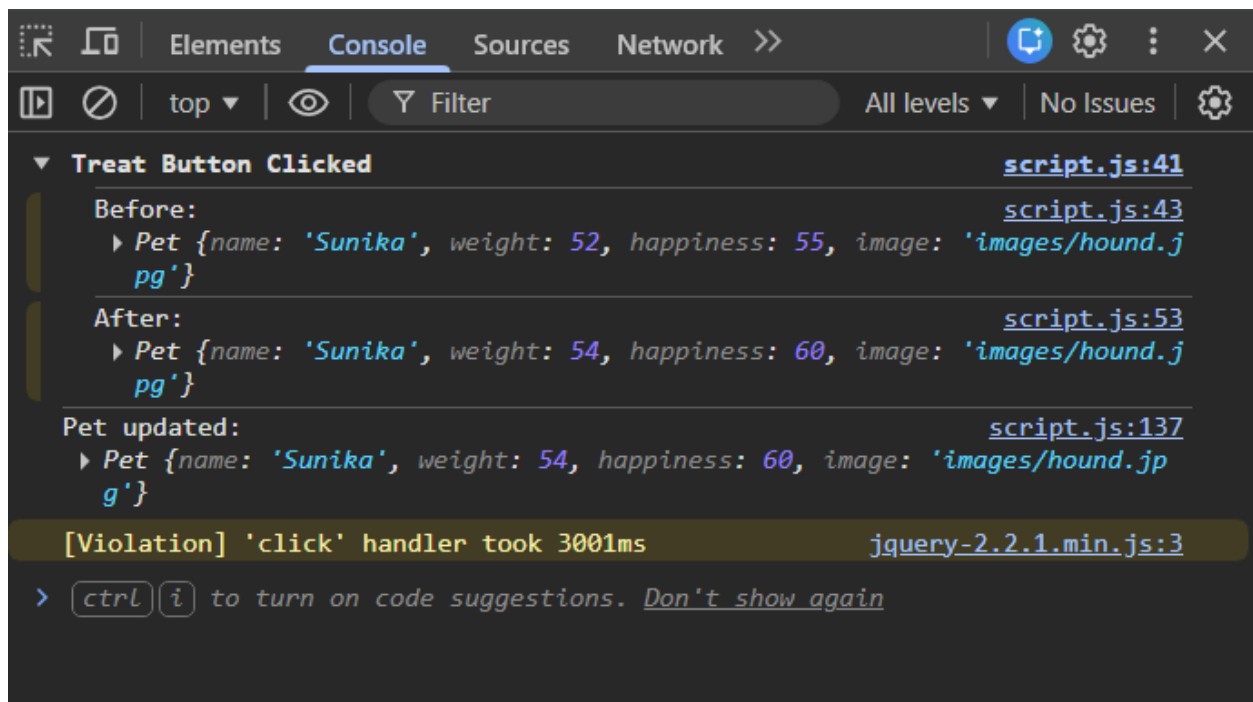



- Cause TypeError

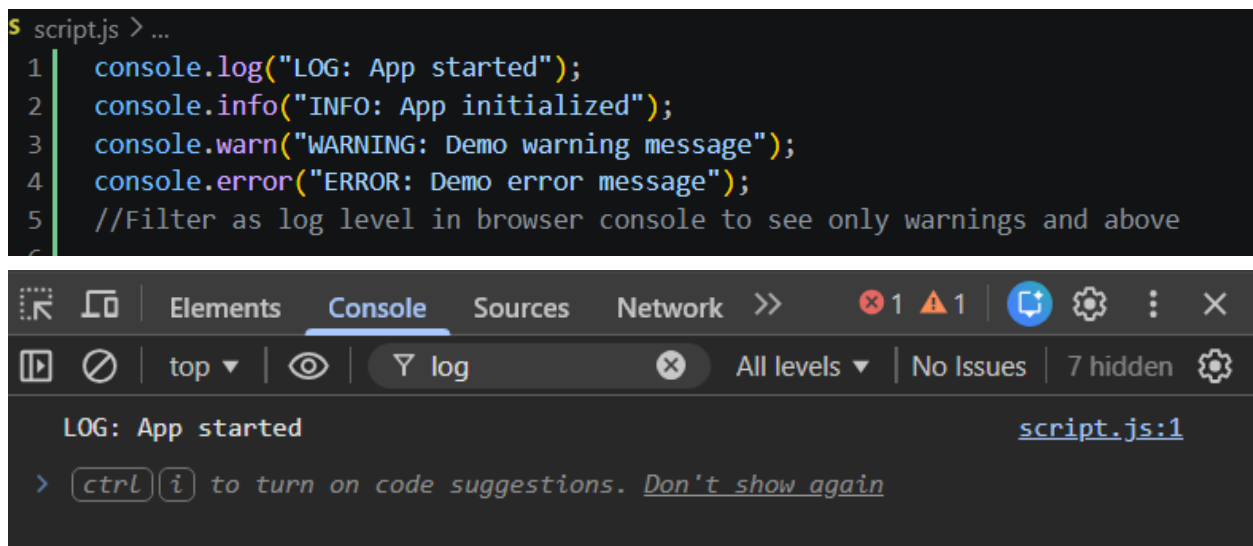


- Cause Violation





- Filter Messages
 - Filter by log level



- Filter by text

```
45 // Treat → +happiness +weight
46 function clickedTreatButton() {
47   console.group("Treat Button Clicked"); // Group logs for better organization
48   console.log("TREAT_EVENT: User gave a treat");
49   console.log("Before:", pet_info); // Log pet info before changes for debugging
50
51   pet_info.happiness += 5;
52   pet_info.weight += 2;
53
54   // Violation Error test: Simulate a long-running operation that blocks the UI
55   //let start = Date.now();
56   //while (Date.now() - start < 3000) {
57   // Busy wait for 3 seconds (blocks UI)
58   //}
59   console.log("After:", pet_info); // Log pet info after changes to verify updates
60
61   console.groupEnd(); // End log group
62
63   showMessage("Yum! That was tasty! 🍖");
64   checkAndUpdatePetInfoInHtml();
65 }
66
67 // Play → +happiness -weight
68 function clickedPlayButton() {
69   console.group("Play Button Clicked");
70   console.log("PLAY_EVENT: User clicked play button");
71   console.log("Before:", pet_info);
```

```
Elements Console Sources Network >>
top Filter All levels No Issues

▼ Play Button Clicked script.js:69
PLAY_EVENT: User clicked play button script.js:70
Before: script.js:71
  ▶ Pet {name: 'Sunika', weight: 51, happiness: 63, image: 'images/hound.jpg'}
After: script.js:76
  ▶ Pet {name: 'Sunika', weight: 50, happiness: 71, image: 'images/hound.jpg'}
Pet updated: script.js:143
  ▶ Pet {name: 'Sunika', weight: 50, happiness: 71, image: 'images/hound.jpg'}

▼ Treat Button Clicked script.js:47
TREAT_EVENT: User gave a treat script.js:48
Before: script.js:49
  ▶ Pet {name: 'Sunika', weight: 50, happiness: 71, image: 'images/hound.jpg'}
After: script.js:59
  ▶ Pet {name: 'Sunika', weight: 52, happiness: 76, image: 'images/hound.jpg'}
Pet updated: script.js:143
  ▶ Pet {name: 'Sunika', weight: 52, happiness: 76, image: 'images/hound.jpg'}
```

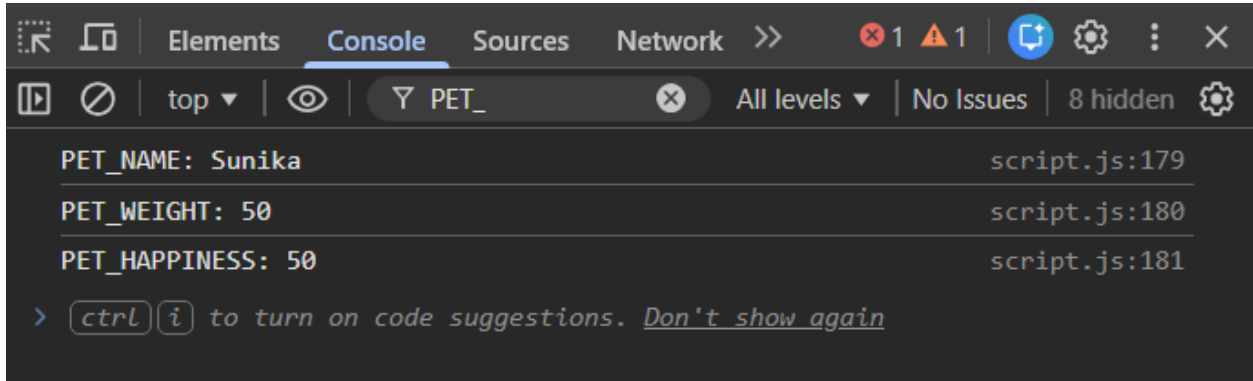
```
Elements Console Sources Network >>
top play All levels No Issues 6 hidden

▼ Play Button Clicked script.js:69
PLAY_EVENT: User clicked play button script.js:70
Before: script.js:71
  ▶ Pet {name: 'Sunika', weight: 51, happiness: 63, image: 'images/hound.jpg'}
After: script.js:76
  ▶ Pet {name: 'Sunika', weight: 50, happiness: 71, image: 'images/hound.jpg'}

> | (ctrl) (i) to turn on code suggestions. Don't show again
```

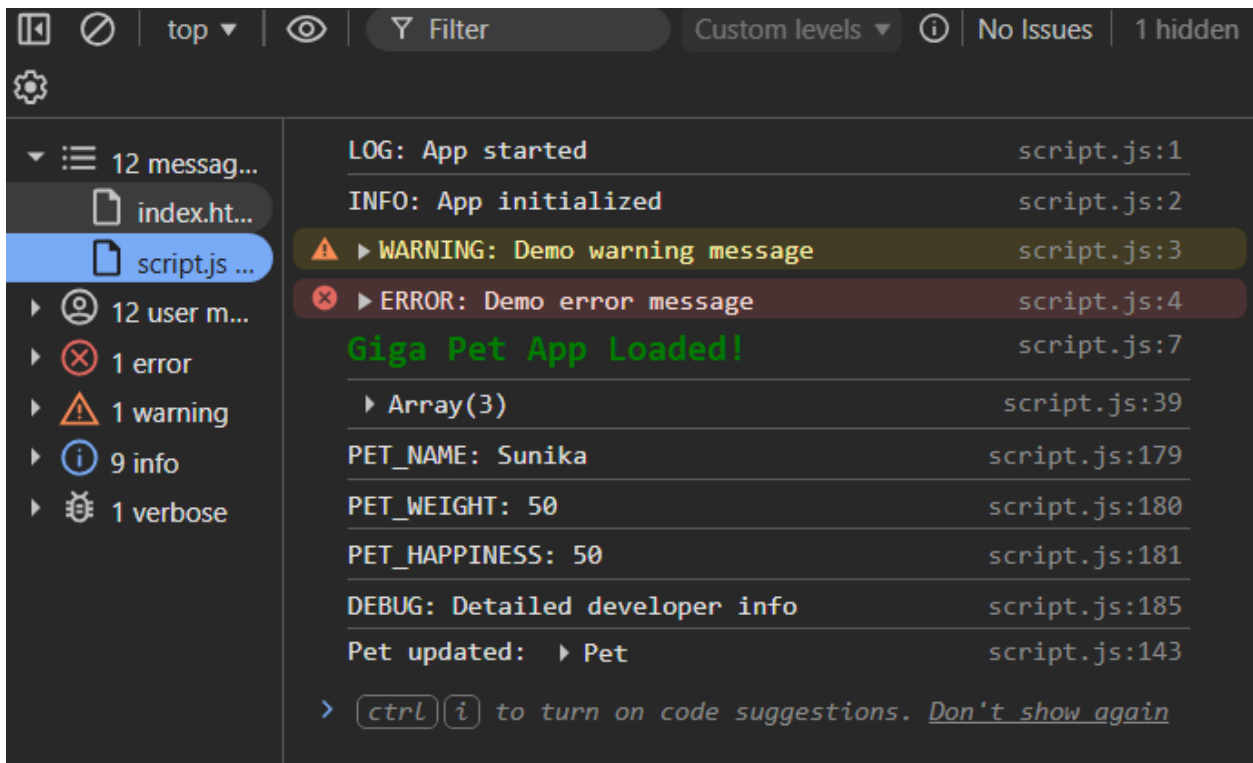
- Filter by regular expression

```
179 console.log("PET_NAME: " + pet_info.name);
180 console.log("PET_WEIGHT: " + pet_info.weight);
181 console.log("PET_HAPPINESS: " + pet_info.happiness);
182 //Filter by regular expression in browser console to see only pet info logs (e.g., "PET_")
183
```



The screenshot shows the browser's developer console with the 'Console' tab selected. A filter 'PET_' is applied to the search bar. The console displays three log messages from 'script.js': 'PET_NAME: Sunika' at line 179, 'PET_WEIGHT: 50' at line 180, and 'PET_HAPPINESS: 50' at line 181. The status bar at the top right indicates 'No Issues' and '8 hidden' messages.

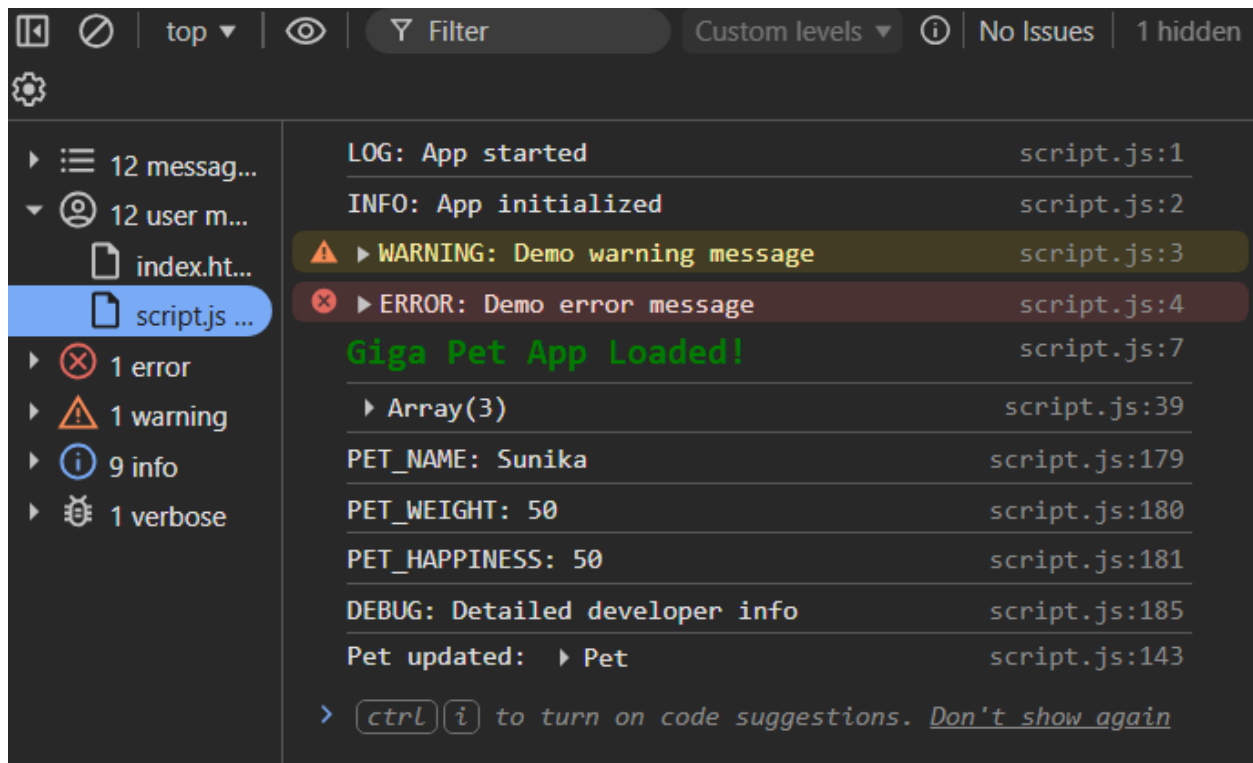
- Filter by message source



The screenshot shows the browser's developer console with the 'Filter' tab selected. The filter bar is empty, and the status bar indicates 'No Issues' and '1 hidden' message. The left sidebar shows a list of messages categorized by type: 12 messages, 1 error, 1 warning, 9 info, and 1 verbose. The main console area displays a list of messages from 'script.js': 'LOG: App started' (line 1), 'INFO: App initialized' (line 2), 'WARNING: Demo warning message' (line 3, highlighted with a yellow background), 'ERROR: Demo error message' (line 4, highlighted with a red background), 'Giga Pet App Loaded!' (line 7, in green), 'Array(3)' (line 39), 'PET_NAME: Sunika' (line 179), 'PET_WEIGHT: 50' (line 180), 'PET_HAPPINESS: 50' (line 181), 'DEBUG: Detailed developer info' (line 185), and 'Pet updated: Pet' (line 143). The status bar at the bottom right indicates 'No Issues' and '1 hidden' message.

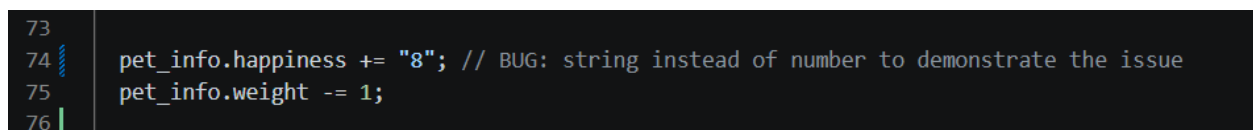
Opened console log and it shows a list of URLs that caused messages to be logged, [script.js](#) shows 11 messages.

- Filter by user messages



Opened user messages on the console log, shows index.html and [script.js](#). [script.js](#) shows 11 messages.

- Reproduce a bug,



The image shows a web application on the left and the Chrome DevTools console on the right.

Web Application:

- Name:** Sunika
- Weight:** 47 pounds
- Happiness:** 50888 tail wags (per min)
- Buttons:** Treat (green), Play (blue), Exercise (red), Sleep (purple), and Switch Pet (orange).
- Feedback:** Two yellow boxes at the bottom each say "That was fun!" with a small dog icon.

Chrome DevTools Console:

- Messages:** 23 messages, 23 user messages, 1 error, 1 warning, 21 info, and No verbose.
- Log Entries:** The console shows three "Play Button Clicked" events from `script.js:70`. Each event includes a `PLAY_EVENT` object with `User clicked play button` and a `Pet` object.
- Pet State Changes:**
 - Before:** `{name: 'Sunika', weight: 50, happiness: 50, image: 'images/hound.jpg'}` (from `script.js:72`)
 - After:** `{name: 'Sunika', weight: 49, happiness: '508', image: 'images/hound.jpg'}` (from `script.js:78`)
 - Pet updated:** `{name: 'Sunika', weight: 49, happiness: '508', image: 'images/hound.jpg'}` (from `script.js:145`)
- Subsequent Events:** Two more "Play Button Clicked" events follow, showing the pet's weight decreasing to 48 and happiness increasing to 5088.
- Tip:** A message at the bottom says: `> [ctrl] i to turn on code suggestions. Don't show again`.

Weight: **47 pounds**

Treat

Play

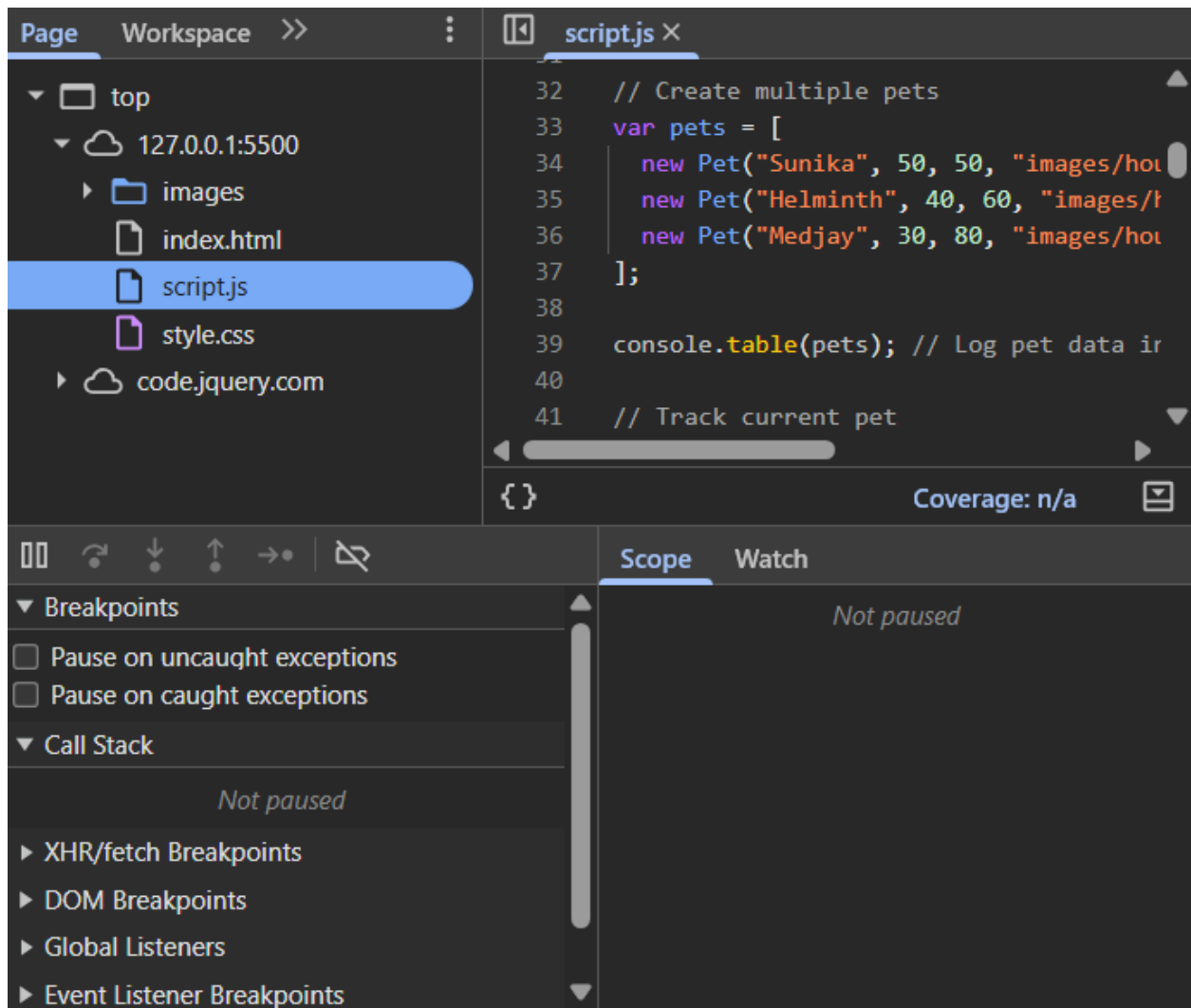
Exercise

Sleep

That was fun! 🏓

That was fun! 🎈

- Get Familiar with the Sources UI,



Elements Console Sources Network Performance Memory >>

Page Workspace >> script.js x

top
127.0.0.1:5500
images
index.html
script.js
style.css
code.jquery.com

```
32 // Create multiple pets
33 var pets = [
34   new Pet("Sunika", 50, 50, "images",
35   new Pet("Helminth", 40, 60, "images",
36   new Pet("Medjay", 30, 80, "images",
37 ];
38
39 console.table(pets); // Log pet data
40
41 // Track current pet
42 var currentPetIndex = 0;
43 var pet_info = pets[currentPetIndex];
44
45 // Treat → +happiness +weight
46 function clickedTreatButton() {
47   console.group("Treat Button Click");
48   console.log("TREAT_EVENT: User gave a treat");
49   console.log("Before:", pet_info);
50
51   pet_info.happiness += 5;
52   pet_info.weight += 2;
53
54   // Violation Error test: Simulate a
55   // console.error("Error: Violation detected");
56 }
```

Watch
Breakpoints
☐ Pause on uncaught exceptions
☐ Pause on caught exceptions
Scope
Not paused
Call Stack
Not paused
XHR/fetch Breakpoints
DOM Breakpoints
Global Listeners
Event Listener Breakpoints
CSP Violation Breakpoints

{ } Coverage: n/a

- Pause the code with a breakpoint

The screenshot displays the Chrome DevTools interface with a breakpoint set on line 74 of `script.js`. The file explorer on the left shows the project structure: `top` (containing `127.0.0.1:5500`), `images`, `index.html`, `script.js`, and `style.css`, along with the `code.jquery.com` directory.

The `Sources` panel shows the following code in `script.js`:

```
62  
63   showMessage("Yum! That was tasty!  
64   checkAndUpdatePetInfoInHtml();  
65   }  
66  
67   // Play → +happiness -weight  
68   function clickedPlayButton() {  
69  
70     console.group("Play Button Clicked");  
71     console.log("PLAY_EVENT: User clicked play button");  
72     console.log("Before:", pet_info);  
73  
74     pet_info.happiness += "8"; // BUG  
75     pet_info.weight -= 1;  
76  
77  
78     console.log("After:", pet_info);  
79  
80     console.groupEnd();  
81  
82     showMessage("That was fun! 🐾");  
83     checkAndUpdatePetInfoInHtml();  
84   }  
85
```

The breakpoint is set on line 74, and the console shows the state of the `pet_info` object before and after the breakpoint is hit:

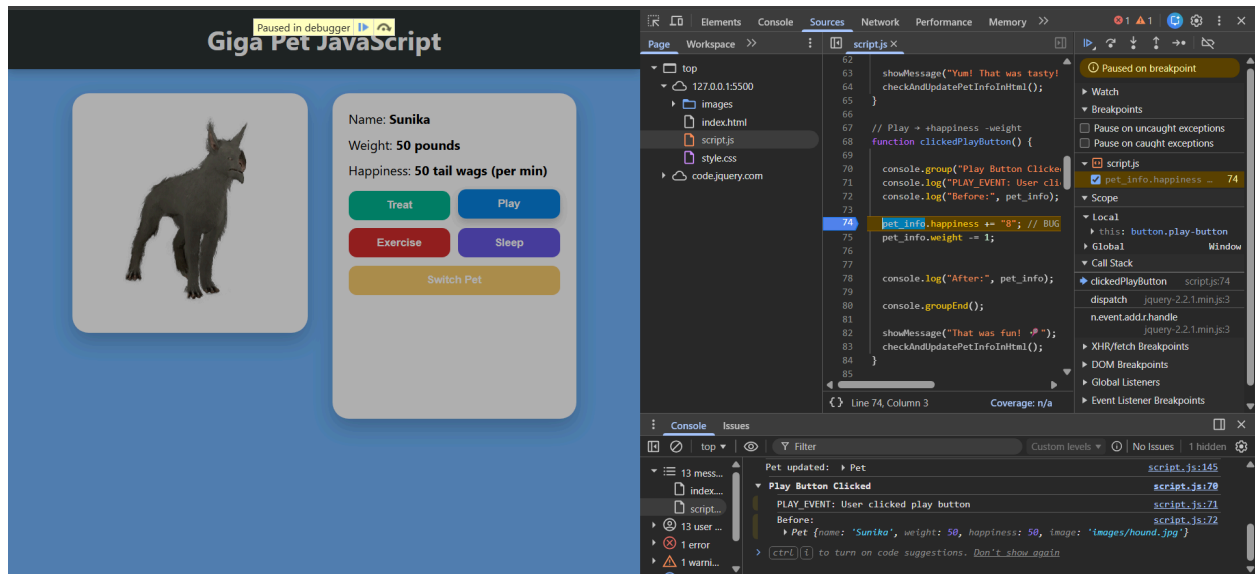
Before: `script.js:72`
▶ Pet {name: 'Sunika', weight: 50, happiness: 50, image: 'images/hound.jpg'}

After: `script.js:78`
▶ Pet {name: 'Sunika', weight: 49, happiness: '508', image: 'images/hound.jpg'}

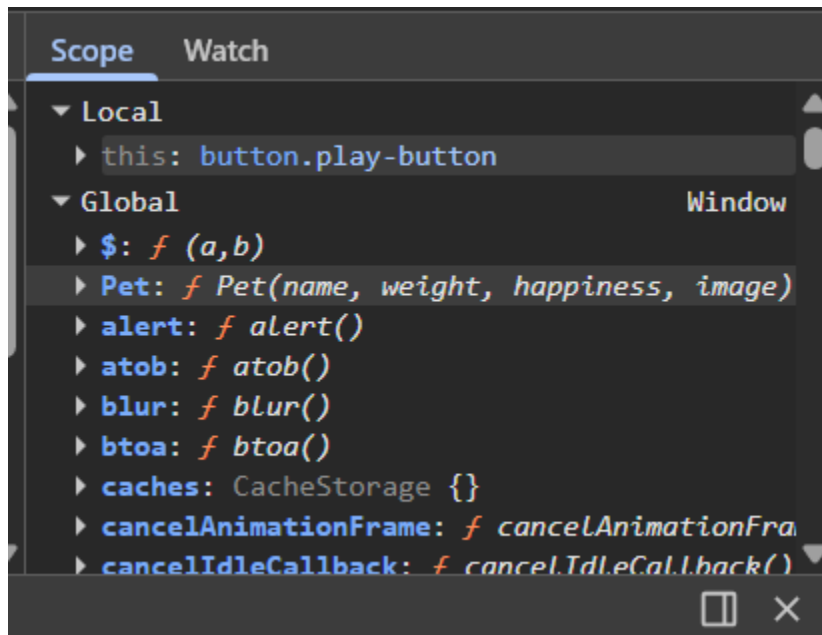
Pet updated: `script.js:145`
▶ Pet {name: 'Sunika', weight: 49, happiness: '508', image: 'images/hound.jpg'}

The console also shows a message: `> (ctrl)(i) to turn on code suggestions. Don't show again`.

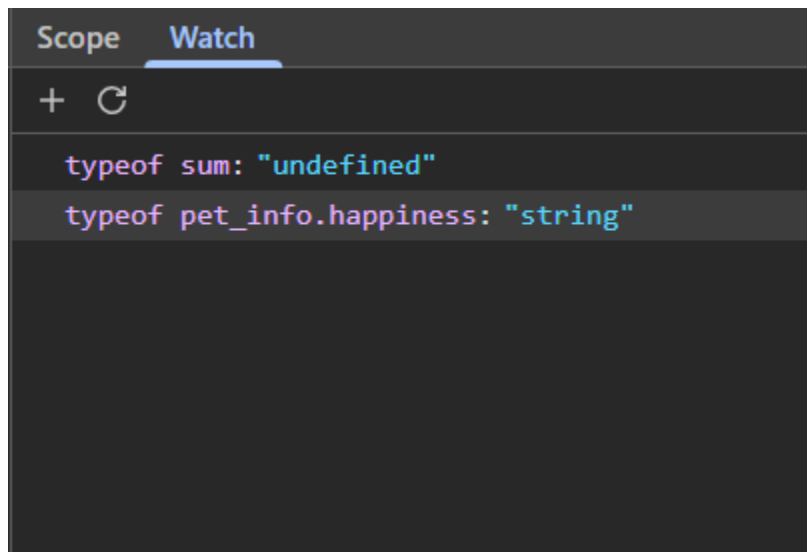
- Set a line-of-code breakpoint,



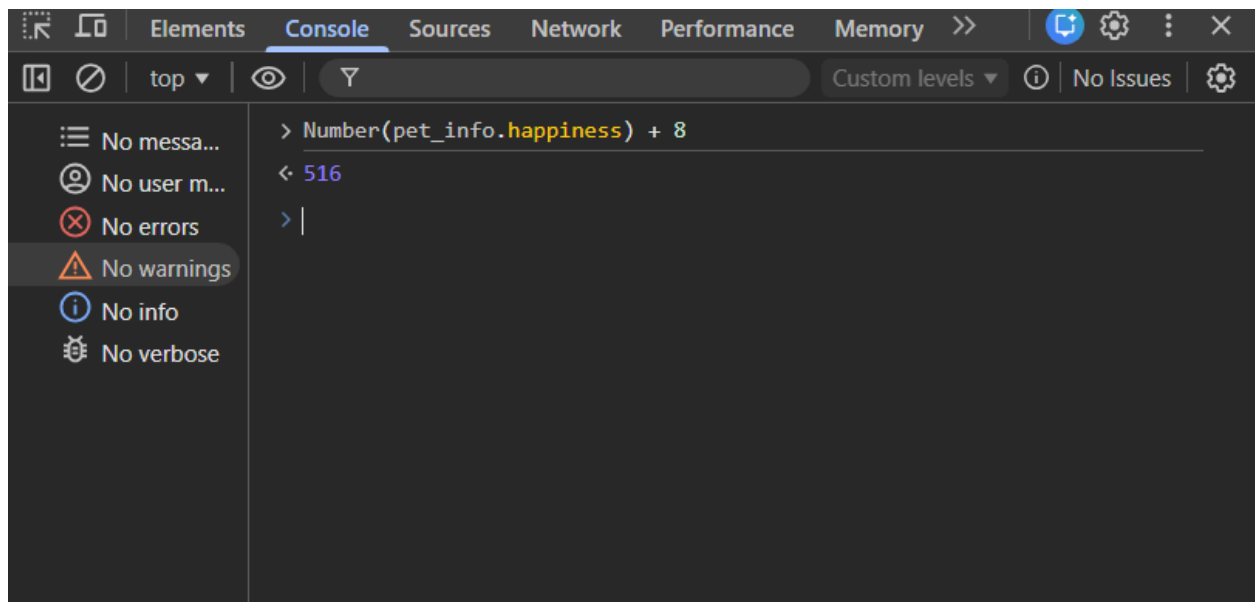
- Check variable values,
 - The Scope Pane



- Watch Expressions,



- The Console,



- Apply a fix

The screenshot shows a web browser's developer tools interface. The top panel displays the 'Sources' tab, showing a file named 'script.js' with a breakpoint set at line 74. The code at line 74 is `pet_info.happiness += 8; // BUG: string instead of`. The bottom panel shows the 'Console' tab, which displays the error message: `TypeError: Cannot read property 'happiness' of undefined`. The 'Watch' tab is also visible, showing the current state of variables: `typeof sum: "undefined"` and `typeof pet_info.happiness: "string"`. The 'Call Stack' shows the function `clickedPlayButton` at line 74 of `script.js`. The 'Console' tab also shows the error message: `TypeError: Cannot read property 'happiness' of undefined`. The 'Watch' tab shows the current state of variables: `typeof sum: "undefined"` and `typeof pet_info.happiness: "string"`. The 'Call Stack' shows the function `clickedPlayButton` at line 74 of `script.js`.

PageWorkspace >>

top

127.0.0.1:5500

images

index.html

*script.js

style.css

code.jquery.com

script.js*

69

70

71

72

73

74

75

76

77

78

79

console.group("Play Button Clicked");

console.log("PLAY_EVENT: User clicked play button");

console.log("Before:", pet_info);

pet_info.happiness += 8; // BUG: string instead of

pet_info.weight -= 1;

console.log("After:", pet_info);

Line 74, Column 25Coverage: n/a

Breakpoints

Pause on uncaught exceptions

Pause on caught exceptions

Call Stack

Not paused

XHR/fetch Breakpoints

DOM Breakpoints

Global Listeners

Event Listener Breakpoints

Scope

Watch

typeof sum: <not available>

typeof pet_info.happiness: <not available>

ConsoleIssues

top

Custom levels

No Issues

3 hidden

3 messages

2 user mes...

No errors

> Number(pet_info.happiness) + 8

< 516